TEACUPS, a python package for simulating time-resolved EPR spectra of spin-polarised multi-spin systems

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Abstract. Spin-polarised magnetic systems, generated by the interaction with light, play a key role in a wide range of scientific applications. Representative examples are OLEDs, photovoltaics, and quantum sensing devices. Further, they are important intermediates in certain biological processes including avian magnetoreception and photosynthesis. Transient electron paramagnetic resonance (trEPR) spectroscopy is a powerful tool to reveal the temporal evolution of non-equilibrium spin states, which contains valuable information on any photoinduced dynamic processes occurring in these systems.

For the analysis of EPR spectra, simulations are essential. However, to date, only static trEPR spectra can be simulated using existing and freely available software and the information contained in the time-evolution is frequently neglected. Here, we introduce `teacups`, a new Python-based simulation routine for time-resolved EPR spectra. By employing the Liouville equation, `teacups` simulates the temporal evolution of trEPR spectra, making it possible to uncover the internal dynamics of spin-polarised systems and to enhance our mechanistic understanding. In this manuscript, we explain the theoretical background and provide a description of the features and setup of `teacups`. Further, a step-by-step example for data analysis using `Teacups` is provided.

1 Introduction

Over the last decades, the study of spin-polarised magnetic systems has emerged as a captivating frontier. These systems, many of which are generated by the interaction with light, are special in the sense that that their energy levels are not populated, leading to population differences far from thermal equilibrium. They play an important role in diverse fields of science and technology. For instance, light-induced triplet states are key intermediates in the processes occurring in certain molecular spintronic devices, organic light-emitting diodes (OLEDs), organic photovoltaics, and can be employed as versatile spin labels.[1–5] Another important research area is the analysis of spin-correlated radical pairs. They are involved e. g. in biological processes such as avian magnetoreception, the circadian clock, and photosynthesis.[6–12] Triplet pair states, generated by singlet fission, are intermediates in the photophysical processes occurring in OLEDs and photovoltaic devices.[13–16] Further, the exploration of the use of paired triplet and doublet states as quantum bits (qubits) in quantum information science is a rather new field of research involving spin-polarised systems.[17–19]

For the analysis of spin-polarised systems, transient electron paramagnetic resonance (trEPR) spectroscopy is a powerful analytical tool.[20] In the experiment, the absorption of microwaves is measured as a function of magnetic field strength and time after pulsed laser excitation. The acquired data reveals the temporal evolution of spin states within non-equilibrium systems and ultimately yields deeper insights into the internal dynamics. This offers a better understanding of the behaviour of coupled spin systems. [20, 21]

For the interpretation of complex EPR spectra simulations are essential. In this work, we introduce the new Python-based simulation routine `teacups` for the analysis of transient EPR spectra. Until now, the simulation of EPR spectra has been limited to static conditions, often performed using the powerful simulation tool EasySpin.[21–23] In our work, such static simulations are extended to encompass the dynamic component. By employing the Liouville equation, time-resolved EPR spectra can be simulated.[24] This offers a better understanding of the temporal evolution of spin-polarised magnetic systems and reveals mechanistic details and internal spin dynamics.

In this manuscript, we present the theoretical background for the simulation followed by a detailed description of the features and setup of the program. We include valuable information for developers as well as for potential users when providing a step-by-step example for an analysis of a trEPR spectrum using the `teacups`-routine and conclude with a short outlook on further developments.

2 Theory

Introduction to transient EPR To understand the physical principles underlying the simulation, some knowledge about the trEPR experiment is necessary as the `teacups` simulation routine generates a transient continuous wave EPR spectrum by replicating the experiment.

Transient EPR is a time-resolved EPR technique which is based on the standard continuous wave (cw) EPR experiment.[20] No field modulation is applied, so that any positive signals correspond to microwave absorption and any negative signals to microwave emission. When running the experiment, the sample is placed into a static magnetic field and exposed to continuous microwave radiation. A nanosecond laser pulse is then applied and a kinetic trace recorded. In the following, the static magnetic field is changed and another kinetic trace is recorded. When repeating this for the entire magnetic field range of interest, a 3D spectrum is obtained, showing the time behaviour as a function of the static magnetic field (see figure 1). The data thus contains information about the spin system as well as the excited state dynamics.[20]

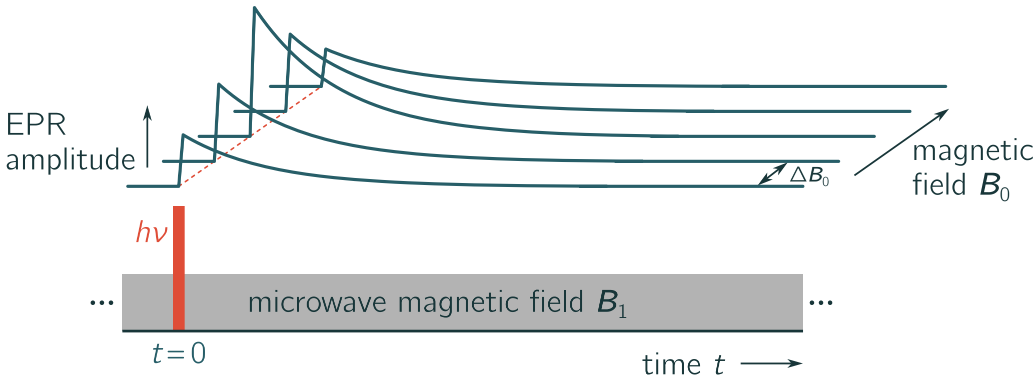

Figure 1: **Schematic drawing of a transient EPR spectrum.** The measurement is started shortly before the application of a short laser pulse. During the experiment, a constant microwave field is applied. At specific magnetic field points, kinetic traces are recorded. By accumulating data for multiple magnetic field points, a 3D data set is obtained.

Mathematical modelling When trying to model the experimental conditions, we have to consider the spin system, the initial polarisation after the laser pulse, and any time-dependent mechanisms involving transitions between the spin states. These three components need to be described mathematically and are fed into the Liouville-von-Neumann equation as shown schematically in figure 2. This equation allows us to calculate the polarisation of the spin system dependent on the time t using multiple operators: The Hamiltonian $\mathcal{H}$, the density operator $\hat{\boldsymbol{\rho}}$, and the time-evolution operator $\hat{\mathbf{R}}$.[24–26] In the following sections, the theoretical construction of a trEPR spectrum is described by explaining the significance of these operators. For this purpose, a simple two-state-system (e.g. a doublet in a magnetic field) is chosen as an example.

Spin system The spin system is defined by the number of spins and their magnetic interactions. Its energy levels are described by the Hamiltonian, which is set up as the sum of the Hamiltonians for all magnetic interactions:

$$\mathcal{H} = \mathcal{H}_{B_0} + \mathcal{H}_{B_1} + \mathcal{H}_{\text{spin-spin}}. \tag{1}$$

The interactions with the *static magnetic field* $\mathcal{H}_{B_0}$ and the time-dependent microwave field $\mathcal{H}_{B_1}$ are defined

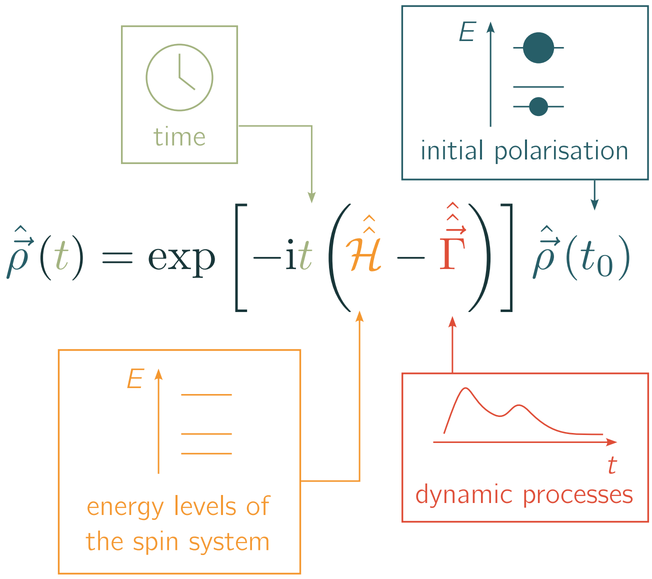

Figure 2: **Liouville-von-Neumann equation.** Multiple input parameters are needed for the calculation of the time-dependent density matrix $\hat{\boldsymbol{\rho}}$. These are: (i) the time point t for which the density matrix shall be calculated, (ii) the Hamiltonian superoperator $\hat{\hat{\mathcal{H}}}$ containing information on the spin system, (iii) the dynamic superoperator $\hat{\hat{\mathbf{R}}}$ containing information on the time-dependent transitions between the states, and (iv) the initial density matrix containing populations and coherences at time point t_0. The resulting density matrix describes the longitudinal and transverse magnetisation of the spin system at the time point t.

by the experimental conditions and are identical for all spin systems. The spin interaction Hamiltonian $\mathcal{H}_{\text{spin-spin}}$ is dependent on the chosen spin system and includes all magnetic interactions between the spins. All operators are set up as matrices. These result from matrix multiplication of the spin matrices (Pauli matrices), the magnetic fields, and the tensors which define the strength and direction of the interactions as explained in the following.

The interaction of the spin system with a static magnetic field is given by the sum of the electronic Zeeman interactions of all spins i with the magnetic field, scaled by the **g**-tensor:

$$\mathcal{H}_{B_0} = \sum_i \hat{\boldsymbol{S}}_i \mathbf{g}_i \boldsymbol{B}_0. \tag{2}$$

where $\hat{\boldsymbol{S}}$ is the spin vector operator $(\hat{\boldsymbol{s}}_x, \hat{\boldsymbol{s}}_y, \hat{\boldsymbol{s}}_z)$ including the Pauli matrices of each spin, and $\boldsymbol{B}_0$ the magnetic field vector which usually lies along the z-axis. Therefore, the equation can be simplified to

$$\mathcal{H}_{B_0} = B_0 \cdot (\hat{\mathbf{s}}_x g_{xz} + \hat{\mathbf{s}}_y g_{yz} + \hat{\mathbf{s}}_z g_{zz}) \tag{3}$$

where the scalar value for the magnetic field strength B_0 and the principal elements of the g-tensor g_i are used.[25, 27, 28]

Since the microwave field oscillates with time, the interaction with the *microwave field* is time-dependent. To simplify calculations, the coordinate system is transferred to the rotating frame by adding a frequency offset to the Zeeman frequencies:

$$\mathcal{H}_{\text{mw}} = \hat{\mathbf{s}}_x g_{\text{iso}} B_1 - \hat{\mathbf{s}}_z \omega_1. \tag{4}$$

The rotating frame rotates with the microwave frequency ω_1 so that the microwave Hamiltonian becomes time-independent. As the microwave field is applied perpendicular to the static magnetic field, the Hamiltonian represents the interaction of the x-part of all spins with the microwave field.

It has to be taken into consideration that all Hamiltonians that are not diagonal will become time-dependent in the rotating frame. As this complicates the calculations immensely, the secular approximation is used for all non-diagonal Hamiltonians, e.g. the dipolar interactions.[27] This is explained in more detail further below in Section 3.

The Hamiltonian for the chosen two-state example system consists only of the interactions of one electron with the magnetic field. The interaction matrix is given by:

$$\mathcal{H} = \begin{pmatrix} \frac{B_0}{2} g_{zz} - \omega_1 & g_{\mathrm{iso}} B_1 \\ g_{\mathrm{iso}} B_1 & -\frac{B_0}{2} g_{zz} + \omega_1 \end{pmatrix}. \tag{5}$$

If the system consists of multiple spins, there are further interactions that must be taken into account. These are the exchange interaction scaled by the exchange interaction parameter J, and the dipolar interactions between the spins scaled by a dipolar interaction tensor $\mathbf{D}$:

$$\mathcal{H} = J\hat{\boldsymbol{S}}_1\hat{\boldsymbol{S}}_2 + \hat{\boldsymbol{S}}_1\mathbf{D}\hat{\boldsymbol{S}}_2. \tag{6}$$

Here, $\hat{\boldsymbol{S}}_1$ and $\hat{\boldsymbol{S}}_2$ are the spin vector operators of two coupled spin systems.[27–29] Common examples are, for instance, two electrons forming a radical pair, the two electrons of a photogenerated triplet state (where the dipolar interaction becomes the zero-field splitting interaction), or a triplet state coupled to a third electron.[4, 21, 30] Further details are given in Section 3.

Polarisation For a light-induced system, as monitored in a trEPR experiment, the populations of the states are not Boltzmann distributed, but depend on the mechanism underlying the formation of the excited species. The resulting spectrum is said to be spin-polarised and typically characterised by the observation of absorptive as well as emissive signals, in contrast to the purely absorptive features observed in the Boltzmann case.[20, 21] The systems' magnetisation is described mathematically by the density matrix operator. The density operator $\hat{\boldsymbol{\rho}}$ is the projection of the states $|\Psi\rangle$ and describes a density distribution. For a two-state system it takes the following general form

$$\hat{\boldsymbol{\rho}} = \sum_i p_i \left|\Psi_i\right\rangle \left\langle\Psi_i\right| = \begin{array}{c|cc} & \text{state 1} & \text{state 2} \\ \text{state 1} & \rho_{11} & \rho_{12} \\ \text{state 1} & \rho_{21} & \rho_{22} \end{array} \tag{7}$$

The diagonal elements represent the probability to find the system in the considered state, i.e. the populations of the states. The population differences result in the longitudinal magnetisation (along the z-axis).[31] If an energetically lower state is overpopulated, this leads to an absorptive spectral feature, whereas an overpopulation of a higher-lying energetic state results in an emissive feature. The off-diagonal elements are the coherences and connect the states. If they are different from zero, this corresponds to a transverse magnetisation (in the xz-plane). At time $t = 0$, the off-diagonal elements are set to zero for the simulation. It is assumed that only population differences are created by the application of the laser pulse. By applying the microwave Hamiltonian, the off-diagonals become populated over time. This creates the detectable transverse magnetisation.[26, 27, 32]

Time-evolution With the Hamiltonian and the density matrix, the system can be fully described at the initial time t_0, as the energetic levels and their populations are defined. In a next step, the time-evolution of the populations must be considered and the resulting transverse magnetisation at every time point has to be calculated. To this end, the full density matrix is required for every time point. The time-dependence of the density operator is described by the Liouville-von-Neumann equation, which can be easily derived from the definition of the density operator:

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\rho}}(t) = -\mathrm{i}\left[\mathcal{H}(t), \hat{\boldsymbol{\rho}}(t)\right], \tag{8}$$

which can be solved by

$$\hat{\boldsymbol{\rho}}(t) = \exp\left[-\mathrm{i}\mathcal{H}t\right]\hat{\boldsymbol{\rho}}(t_0)\exp\left[\mathrm{i}\mathcal{H}t\right]. \tag{9}$$

Here, the full Hamiltonian and the density matrix at t_0 are needed as input parameters. Changes in the density matrix due to the microwave field are taken into consideration as part of the Hamiltonian.[23, 24, 26, 33, 34] However, any further dynamic changes, like, for instance, relaxation processes, cannot be taken into account using this equation that emerged as a solution directly from the differential equation. In order to take such processes into account, the density matrix operator itself would have to be changed. But operator changes are not possible in Hilbert space.

Dynamic processes can be taken into account when changing from the Hilbert space to the Liouville space. In the Hilbert space operators act on functions, whereas in Liouville space (i.e. a space "above") superoperators (marked with a double-hat) act on operators. Here, each matrix operator with the dimension $n \times n$ is written as a vector with n^2 elements. These vectors are transformed by the superoperators, which are matrices with the dimension $n^2 \times n^2$. Hence, a superoperator that influences the density operator can be set up.

When changing from Hilbert to Liouville space, the Lioville-von-Neumann equation is transformed to the master equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\rho}}(t) = -\mathrm{i}\left(\hat{\hat{\mathcal{H}}} + \mathrm{i}\hat{\hat{\boldsymbol{R}}}\right)\hat{\boldsymbol{\rho}}(t)\,, \tag{10}$$

which is solved by

$$\hat{\boldsymbol{\rho}}(t) = \exp\left[-\mathrm{i}t\left(\hat{\hat{\mathcal{H}}} + \mathrm{i}\hat{\hat{\boldsymbol{R}}}\right)\right]\hat{\boldsymbol{\rho}}(t_0) \tag{11}$$

where $\hat{\hat{\boldsymbol{R}}}$ is the dynamic superoperator which describes transitions between the different states of the system. Further superoperators could still be added. The Hamiltonian superoperator includes the commutator relation from equation 8.[24, 26, 31, 34, 35]

To illustrate how to construct the dynamic superoperator, it will be analysed here for the two-state example system. As the operator dimension is 2×2, the superoperator has a dimension of 4×4:

$$\hat{\hat{\boldsymbol{R}}} = \begin{array}{c|cccc} & 11 & 12 & 21 & 22 \\ 11 & k_{11,11} & k_{11,12} & k_{11,21} & k_{11,22} \\ 12 & k_{12,11} & k_{12,12} & k_{12,21} & k_{12,22} \\ 21 & k_{21,11} & k_{21,12} & k_{21,21} & k_{21,22} \\ 22 & k_{22,11} & k_{22,12} & k_{22,21} & k_{22,22} \end{array} \tag{12}$$

Here, the indices ij, kl describe which elements ij and kl of the density matrix are affected by the rate constant $k_{ij,kl}$. Density is transferred from the element ij to the element kl. Like this, modulations can be defined using the dynamic superoperator.[24, 30, 34, 36] The following rules apply:

- a positive k defines an increase of an element; a negative k a decrease
- k is always a rate constant that defines a proportionate change of the current value
- changes in populations are described by the elements with $i = j$ and $k = l$
- changes in coherences are described by the elements with $i \neq j$ and $k \neq l$
- the total spin density is only preserved if $k_{ij,kl} = -k_{ij,ij}$

With the information given above, the density matrix can now be calculated for every time point. The only remaining step is thus the calculation of the EPR signal itself. In transient continuous wave EPR the detection is carried out perpendicular to the static magnetic field. Consequently, the spin operator along the y-axis is chosen as the detection operator. To obtain the EPR signal, the expectation value of the detection operator is calculated,[27, 33] which is defined as

$$\begin{aligned}\text{Hilbert space: } \langle \hat{\boldsymbol{s}}_y \rangle &= \mathrm{Tr}\left[\hat{\boldsymbol{\rho}}\,(t)\,\hat{\boldsymbol{s}}_y\right] \\ \text{Liouville space: } \langle \hat{\boldsymbol{s}}_y \rangle &= \hat{\boldsymbol{\rho}}\,(t)\,\hat{\boldsymbol{s}}_y.\end{aligned} \tag{13}$$

3 Simulations

In the following section the specific features of `teacups` shall be outlined and explicit matrices and formulas will be given. `teacups` is an open source Python package, that can be used by script. The script form gives maximal flexibility to the user; spin system and dynamics can be defined easily. A modular setup of the functions and a good documentation make it possible for users with some experience in Python to extend the program themselves by adding own functions, e.g. for further spin systems, different initial populations or polarisation transfer mechanisms. Details on the programming are given in the Supporting Information.

As explained in the theory section, the simulation input consists of three main parts: The spin system, the initial polarisation, and the dynamic evolution, all together resulting in the simulated spectrum. For each part, `teacups` provides a selection of typical scenarios, that can be combined with each other. This makes the program easy to use, while further components can be added with knowledge of theory and programming. The available presets and their mathematical descriptions are presented in the following.

Box 2: Basis and basis transformation[31, 37]
All matrices and vectors that are to be offset against each other must be in a vector space with the same basis. The basis of a vector space consists of the minimum number of linearly independent vectors. Each vector in the vector space can then be represented as a linear combination of these basis vectors. The simplest example for a two-dimensional space are the unit vectors.
For spectral simulations, it makes sense to use different bases depending on the situation. Examples are the eigenbasis of the Hamiltonian, in which it is diagonal, or the singlet-triplet basis for a radical pair, in which the initial occupation of the density matrix can be easily specified. Since the operators are partly set up in different bases, a basis transformation must be carried out in order to transfer all vectors and matrices into the same basis. The basis transformation matrix $\boldsymbol{T}_{B'}^{B}$ transferring a matrix from the old basis B to the new basis B' is set up by representing the vectors of the old basis as a linear combination of the vectors of the new basis. The basis transformation matrix is then applied to the vectors $\boldsymbol{V}$ and matrices $\boldsymbol{M}$

$$\begin{aligned}\boldsymbol{V}\,(B') &= \boldsymbol{T}_{B'}^{B}\boldsymbol{V}\,(B) \\ \boldsymbol{M}\,(B') &= \boldsymbol{T}_{B'}^{B}\boldsymbol{M}\,(B)\,\boldsymbol{T}_{B}^{B'}.\end{aligned}$$

Powder average In all simulations using `teacups`, powder spectra are simulated. Hence, for an anisotropic system, all orientations have to be considered. To do so, a powder average is constructed.

All tensors (after rotating them into a common initial frame) are rotated by a set of Euler angles using the rotation matrix as a basis transformation matrix[38]

$$\hat{\boldsymbol{r}} = \begin{pmatrix} \cos\phi\cos\vartheta & -\sin\phi & \cos\phi\sin\vartheta \\ \sin\phi\cos\vartheta & \cos\phi & \sin\phi\sin\vartheta \\ -\sin\vartheta & 0 & \cos\vartheta \end{pmatrix} \tag{14}$$

The set of angles consists of several pairs of ϑ and ϕ, describing points that are equally distributed on a unity sphere. For this purpose, a SOPHE[22] grid and a fibonacci grid[39] are implemented. When rotating all tensors to a particular orientation, a spin system in this orientation is described. By calculating spectra for a large number of orientations and summing them up with equal weights, a powder averaged spectrum is obtained.[27]

Implemented spin systems Predefined Hamiltonians for four spin systems (see Figure 3) that are often investigated using trEPR are provided. These are a doublet (`doub`), a spin-correlated radical pair (`rp`), a triplet state (`trip`), and a triplet-doublet pair (`tdp`). Each species has its own set of magnetic interactions. The Hamiltonian matrices are shown in the Supporting Information.

The *doublet* consists of a single spin in a magnetic field. The spin quantum number is $s = 1/2$. In a magnetic field, two distinct energetic states exist and one transition is expected. The Hamiltonian includes only the Zeeman interaction with the magnetic field

$$\mathcal{H}_{\texttt{doub}} = \hat{\mathbf{s}}_{1/2}\mathbf{g}B_0. \tag{15}$$

As a basis for all calculations, the two eigenfunctions α and β are chosen, which results in a diagonal Hamiltonian.[27]

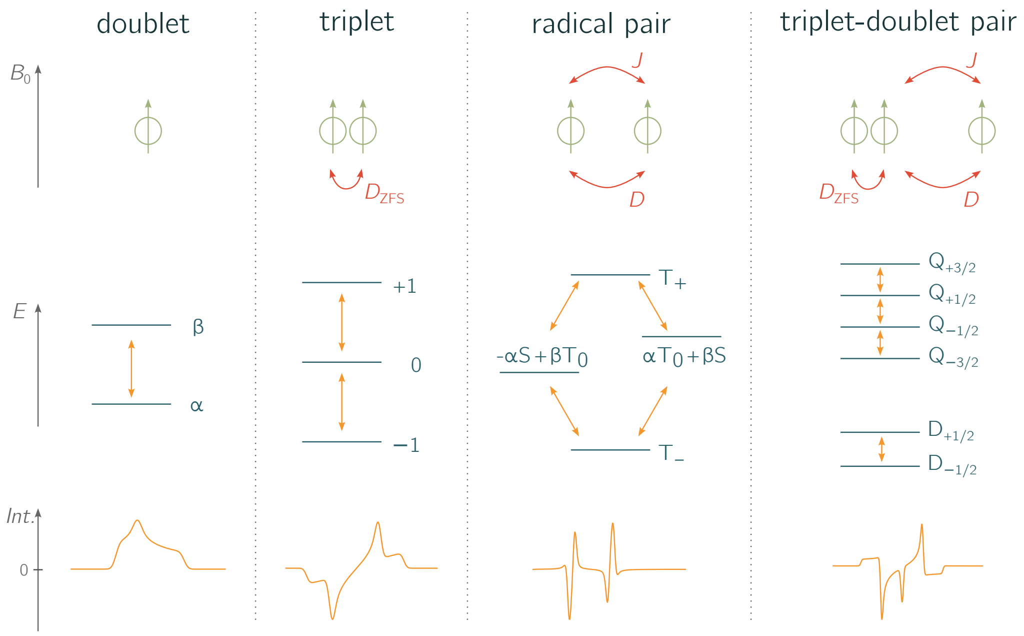

Figure 3: **Overview of the implemented spin systems.** The four spin systems that are currently implemented in `teacups` are shown. For each spin system, the involved spins and their interactions are sketched (top). Further, the eigenstates in a magnetic field and the allowed transitions between them are illustrated (centre) together with a typical spin-polarised spectrum (bottom).

The *radical pair* consists of two spins $s_{1,2} = 1/2$ that are coupled, which means that interactions between the two spins exist. This leads to a system of four spin states with four allowed transitions between them. The distance between the two electrons in a spin-correlated radical pair is such that exchange and dipolar interactions need to be taken into account. Both are considered in the Hamiltonian which takes the form

$$\mathcal{H}_{\texttt{rp}} = \hat{\mathbf{s}}_1\mathbf{g}_1B_0 + \hat{\mathbf{s}}_2\mathbf{g}_2B_0 + J\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2 + \hat{\mathbf{s}}_1\mathbf{D}\hat{\mathbf{s}}_2. \tag{16}$$

Instead of the product basis, where the functions are equal to the products of the radical eigenstates α and β, the singlet-triplet basis is chosen

$$\begin{aligned}
|T_+\rangle &= |\alpha,\alpha\rangle \\
|S\rangle &= \frac{1}{\sqrt{2}}\left(|\alpha,\beta\rangle - |\beta,\alpha\rangle\right) \\
|T_0\rangle &= \frac{1}{\sqrt{2}}\left(|\alpha,\beta\rangle + |\beta,\alpha\rangle\right) \\
|T_-\rangle &= |\beta,\beta\rangle\,.
\end{aligned} \tag{17}$$

In this basis, the Hamiltonian is (nearly) diagonal and can be assumed to be time-independent in the rotating frame.[21, 30]

The *triplet* state consists of two spins in very close proximity. As a consequence, the exchange interaction, and therefore also the energetic gap between the singlet and triplet states, becomes very large. The total system can be described by the three triplet functions, neglecting the singlet function. Hence, the total spin of a triplet is $s = 1$. Three triplet states exist and are connected by two allowed transitions. They are split by a dipolar interaction at zero field (zero-field-splitting, ZFS). The magnitude of the ZFS is defined by the two ZFS-parameters D and E. The corresponding Hamiltonian is given by

$$\begin{aligned}
\mathcal{H}_{\mathtt{trip}} &= \hat{\boldsymbol{s}}\mathbf{g}B_0 + \hat{\boldsymbol{s}}\mathbf{D}_{\mathrm{ZFS}}\hat{\boldsymbol{s}} \\
&= \hat{\boldsymbol{s}}\mathbf{g}B_0 + D\left(\hat{\boldsymbol{s}}_z^2 - \frac{1}{3}\hat{\boldsymbol{s}}^2\right) + E\left(\hat{\boldsymbol{s}}_x^2 - \hat{\boldsymbol{s}}_y^2\right).
\end{aligned} \tag{18}$$

When applying the secular approximation, only the z-components of the $\mathbf{D}$-tensor are taken into account,[27] making the Hamiltonian time-independent in the rotating frame. As a basis for the triplet simulation, the eigenfunctions of the triplet in high field, $|T_+\rangle$, $|T_0\rangle$ and $|T_-\rangle$, are chosen.[4, 21]

As the name implies, a *triplet–doublet pair* consists of a coupled triplet ($s = 1$) and a doublet ($s = 1/2$) state. This results in six eigenstates. Both the dipolar and exchange interactions need to be taken into account and are included in the Hamiltonian. The latter is similar to the Hamiltonian of the radical pair, except for the ZFS interaction that needs to be included for the triplet state

$$\mathcal{H}_{\mathtt{tdp}} = \hat{\boldsymbol{s}}_{\mathrm{doub}}\mathbf{g}_{\mathrm{doub}}B_0 + \hat{\boldsymbol{s}}_{\mathrm{trip}}\mathbf{g}_{\mathrm{trip}}B_0 + \hat{\boldsymbol{s}}_{\mathrm{trip}}\mathbf{D}_{\mathrm{ZFS}}\hat{\boldsymbol{s}}_{\mathrm{trip}} + J\hat{\boldsymbol{s}}_{\mathrm{doub}}\hat{\boldsymbol{s}}_{\mathrm{trip}} + \hat{\boldsymbol{s}}_{\mathrm{doub}}\mathbf{D}\hat{\boldsymbol{s}}_{\mathrm{trip}} \tag{19}$$

The magnitude of the exchange interaction defines the number of allowed transitions. For a strongly coupled pair, the quartet and doublet functions include four allowed transitions as shown in the figure. For moderately or weakly coupled systems, the energy differences between the eigenfunctions are small and additional transitions become allowed. As a basis, the product basis is chosen (products of the high-field triplet eigenfunctions and the doublet eigenfunctions). The secular approximation is applied for the dipolar coupling and the ZFS to make the Hamiltonian time-independent.[40–42]

Initial polarisation With a good knowledge of the theoretical concepts, it is possible for the user to provide a custom initial density matrix to the `teacups` simulation routine. The most important aspect to consider is the chosen basis. For convenience, some initial polarisations (hereafter initial states) are already implemented and can be selected for the simulation.

The user can choose one out of seven bases and set the initial populations of the states. For all implemented initial states, it is assumed that, initially, only longitudinal magnetisation exists, i.e. coherences are neglected in the initial density matrices.

If `'basis'` is chosen as the initial state, the density matrix is set up directly in the basis of the system: The basis of the input density matrix is the same as the basis for the calculations. The population input is set as the diagonal elements of the matrix and directly given to the simulation routine:

$$[a, b, \cdots, n] \rightarrow \hat{\boldsymbol{\rho}}_0 = \begin{pmatrix} a & 0 & \cdots & 0 \\ 0 & b & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n \end{pmatrix} \tag{20}$$

If the high field states of the system are to be populated, the initial state `'eigen'` has to be chosen. Here, the input populations are set as the diagonal elements of a matrix in a basis spanned by the system's eigenstates for every magnetic field point. Internally, the matrix is then transformed to the basis of the Hamiltonian by basis transformation with the basis transformation matrix U:

$$\mathcal{H}_{\text{diagonal}} = U\mathcal{H}U^{-1} \tag{21}$$

yielding a diagonal Hamiltonian of the form

$$\hat{\boldsymbol{\rho}}_0 = U^{-1} \begin{pmatrix} a & 0 & \cdots & 0 \\ 0 & b & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n \end{pmatrix} U \tag{22}$$

Most EPR spectra of spin-polarised triplet states can be described well when considering an initial population of the zero-field triplet eigenstates. In the program, the zero-field population can be selected by the user as the basis for a triplet state simulation by choosing the `'zf'`-initial state. Here, the populations that need to be given to the simulation routine are the populations of the three zero-field states of the triplet, T_X, T_Y and T_Z, in ascending order (with reference to their energies). The matrix is transformed afterwards to the high-field basis T_+, T_0 and T_- for each magnetic field point by basis transformation with the basis transformation matrix U[37, 43]

$$\mathcal{H}_{\text{diagonal}} = U\mathcal{H}_{\text{triplet, high-field}}U^{-1} \tag{23}$$

yielding the diagonal Hamiltonian

$$\hat{\boldsymbol{\rho}}_0' = U^{-1} \begin{pmatrix} a & 0 & \cdots & 0 \\ 0 & b & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n \end{pmatrix} U, \tag{24}$$

To obtain $\hat{\boldsymbol{\rho}}_0$, all off-diagonal elements, i.e. coherences, are set to zero after the basis transformation.[21]

For the coupled systems, namely the radical pair and the triplet–doublet pair, further initial states can be chosen. Radical pairs may be generated from a singlet or a triplet precursor. If `'singlet'` is chosen as the initial state, only the singlet state is populated and the population of all triplet states is set to zero. The initial density matrix is set up in the singlet–triplet basis according to

$$\hat{\boldsymbol{\rho}}_0 = \begin{array}{c} \begin{array}{cccc} \mathrm{T}_+ & \mathrm{S} & \mathrm{T}_0 & \mathrm{T}_- \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}. \tag{25}$$

To simulate a radical pair with a triplet precursor, the three populations of the triplet states have to be given. Here, it is possible to give the high-field populations (`'triplet-eigen'`) or zero-field populations (`'triplet-zf'`). The density matrix is set up as explained above populating the triplet levels of the radical pair density matrix.[43]

For the triplet–doublet pair, the density matrix is set up using two sets of populations: The population of the doublets and that of the triplets. Here again `'triplet-eigen'` or `'triplet-zf'` can be chosen, which defines the basis for the triplet. The doublet is set up in its eigenbasis states α and β. As the Hamiltonian is set up in the product basis, the density matrix is also set up in the product basis of high-field triplet and doublet functions. To this end, the Kronecker product is computed[38]

$$\hat{\boldsymbol{\rho}}_0 = \hat{\boldsymbol{\rho}}_{\text{triplet}} \otimes \boldsymbol{E} + \boldsymbol{E} \otimes \hat{\boldsymbol{\rho}}_{\text{doublet}}. \tag{26}$$

Alternatively, the `'coupled'` basis can be chosen. In this case, the density matrix is set up in the basis of a strongly coupled triplet–doublet pair, i.e. the doublet and quartet states are populated (analogous to the population of singlet and triplet states in case of a radical pair). After the initial setup, the density matrix is internally transformed to the product basis for further calculations.[40]

Implemented relaxation mechanisms To simulate the time-evolution, relaxation models of different complexity can be chosen. First of all, the user needs to choose the space that is used for the calculations. This can either be `'hilbert'` or `'liouville'`.

If the Hilbert space is chosen, the advantage is that the operator dimensions are smaller. Therefore, calculations are much faster than in Liouville space. However, while the static EPR spectrum is simulated correctly, no dynamic superoperators can be applied. As a consequence, only a system-intern time-evolution in form of transient nutations can be simulated but additional dynamic processes cannot be accounted for.[31] To simulate a phenomenological decay, the spectrum is convoluted with an exponential decay of the form

$$S_{\text{decay}} = S * \exp\left[-\tau t\right]. \tag{27}$$

characterised by the decay time constant τ.

If `'liouville'` is chosen, the operators are transformed to superoperators and vectors as described in the theory part. This squares their dimensions and slows down the computation, but dynamic superoperators can be included.[31] Regarding the dynamic superoperator, different inputs are possible. The simplest way to simulate relaxation is by using the longitudinal relaxation time T_1 and the transverse relaxation time T_2. This describes a phenomenological decay of the signal but the model is more complex than a simple exponential decay and the decay times have a physical meaning: The longitudinal relaxation time drives the populations to equilibrium, while the transverse relaxation time describes the decay of the coherences. The relaxation superoperator is implemented using the relaxation times T_1 and T_2. The corresponding matrix can be found in the Supporting Information.[24, 26, 30, 36]

Alternatively, for complete flexibility, the user can define their own dynamic superoperator. Here, it is necessary to have some prior knowledge about the spin system and the feasibility of certain dynamic processes since a specific model needs to be chosen. Transitions between spin states can be initialised by defining the relevant matrix elements as described in the theory part.

Simulation example In the following section, an example of how to use `teacups` is provided. The script used for the simulation can be found in the Supporting Information. The steps from an experimental time-resolved spectrum to the final 3D-simulation are explained and are visualised in figure 4.

As an example, the transient EPR spectrum of a perylene–nitroxide conjugate, referred to as perylene-3-*e*TEMPO, is given. It is a covalently bound, photogenerated triplet–doublet system consisting of a stable radical

Figure 4: **From an experimental spectrum to a teacups simulation.** The simulation of a time-resolved EPR spectrum requires four steps which are illustrated here on the example of a photoexcited perylene–nitroxide molecule forming a triplet–doublet pair. The spin system parameters are determined by a quick Hilbert-space simulation of the EPR spectrum at early times after photoexcitation. The doublet and quartet eigenstates are plotted. Two rate constants, for intersystem crossing (isc) and internal conversion (ic), were chosen here to describe relaxation processes between the states. The whole spectrum and the evolution of the populations is finally plotted. Important simulation options are indicated in grey and the resulting plots are shown for each step.

and a chromophore. After photoexcitation at 435 nm, the chromophore triplet state is formed rapidly by a process referred to as radical-enhanced intersystem crossing. Since, triplet state and radical doublet state are strongly coupled, four quartet states and two doublet states are formed. The experimental data set that is to be described is typical for a quartet state. The spectrum shows a broad multiplet polarisation arising from transitions between the quartet state $Q_{\pm 3/2} \leftrightarrow Q_{\pm 1/2}$ levels and a sharp net polarisation in the centre of the spectrum arising from transitions between the quartet state $Q_{+1/2} \leftrightarrow Q_{-1/2}$ levels. When examining the experimental time-evolution of the signals, at least two different kinetic processes can be identified: The net polarisation decays much faster than the multiplet polarisation.[19] Such a spectrum can be analysed using `teacups`. To this end, the following workflow may be used:

1. *Characterisation of the spin system:* An EPR spectrum at early times after photoexcitation is chosen from the 2D data set which clearly shows all initial features. For a quick simulation, the time points are reduced to two and the Hilbert space is chosen. The resulting spectrum is then plotted. If required, the initial spin system parameters can be adjusted until a satisfactory agreement between experiment and simulation is reached. Further, an appropriate basis needs to be chosen, which may require a readjustment of the initial polarisation.

Now, the starting conditions are set.

2. *Analysis of the energy of the eigenstates:* Now the eigenstates of the system can be verified. Upon request `teacups` returns a plot of the energy levels of the eigenstates as a function of the magnetic field. In the case of our example, we can see here that the four quartet states are well separated from the two doublet states. From this plot, the energetic order of the states can be determined, which is needed to set up the matrix describing the dynamic processes.

3. *Identification of plausible relaxation mechanisms:* Intersystem crossing (isc) and internal conversion (ic) are plausible mechanisms leading to a decay of the signal and a redistribution of population. As a simple model, two rate constants k_{isc} and k_{ic} can be assumed. Now the respective states are connected with the rate constants in a dynamic matrix.

4. *Calculation of the spectrum and the time-evolution of the eigenstate populations:* The spectrum and the evolution of the populations can now be calculated. To this end, the Lioville space is chosen in combination with a sufficient number of time points. The rate constants are adjusted until the simulated signal decay fits the experimental data.

4 Summary and Outlook

With `teacups`, a Python simulation routine for the simulation of time-resolved EPR spectra of spin-polarised species has been developed. It is possible to simulate powder spectra of doublets, triplets, spin-correlated radical pairs, and coupled doublet–triplet pairs. The spin system parameters can be chosen as well as the initial populations of the states. Further, a flexible matrix describing the dynamics of the system can be defined. After simulation of the spectrum, the evolution of the populations between the eigenstates can be compared with the experimental data. This makes it possible to analyse the effect of population-transfer on the spectrum or to verify a plausible dynamic mechanism.

`teacups` has a modular setup that guarantees maximal flexibility but has also some predefined models, that are easy to use. The modular setup allows the integration of further models. These could be further spin systems like, for instance, triplet–triplet pairs or a doublet coupled to one or multiple nuclei,[44] further dynamic processes (e.g. the reverse quartet mechanism)[45] or further initial state matrices (e.g. with initially populated coherences).

At the present stage, it is still necessary to make the program more efficient. Some efforts have already been made in this regard and are described in the SI. If the simulation routine works faster and uses less memory, further simulation parameters could be included, such as hyperfine interactions.

A further interesting feature to be added could be the transition between two different spin systems. To this end, a full Hamiltonian consisting of the combination of both systems would need to be set up. Then, a superoperator matrix for the transition between all states can be defined. Functions of `teacups` for the build-up and propagation of the signal could be used for this purpose without changes.

Finally, we can conclude that the presented simulation routine has great potential to develop into a flexible open-source program for the simulation of time-resolved EPR spectra. This has been realised by providing a simple framework for the calculation of spin-polarised EPR spectra based on the density matrix formalism.

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5 Supporting information

5.1 Explicit matrices

5.1.1 Hamiltonians

$$\mathcal{H}_{\text{doub}} = \begin{pmatrix} \frac{B_0}{2} g_{zz} & 0 \\ 0 & -\frac{B_0}{2} g_{zz} \end{pmatrix} \tag{28}$$

$$\mathcal{H}_{\text{trip}} = \begin{pmatrix} B_0 g_{zz} + \frac{D_{zz}}{2} & 0 & 0 \\ 0 & -D_{zz} & 0 \\ 0 & 0 & -B_0 g_{zz} + \frac{D_{zz}}{2} \end{pmatrix} \tag{29}$$

$$\mathcal{H}_{\text{rp}} = \begin{pmatrix} \frac{B_0}{2}\left(g1_{zz} + g2_{zz}\right) - J - \frac{D_{zz}}{2} & 0 & 0 & 0 \\ 0 & J & \frac{B_0}{2}\left(g1_{zz} - g2_{zz}\right) & 0 \\ 0 & \frac{B_0}{2}\left(g1_{zz} - g2_{zz}\right) & -J + D_{zz} & 0 \\ 0 & 0 & 0 & -\frac{B_0}{2}\left(g1_{zz} + g2_{zz}\right) - J - \frac{D_{zz}}{2} \end{pmatrix} \tag{30}$$

$$\begin{aligned} \mathcal{H}_{\text{tdp}} = {} & \mathcal{H}_{\text{doub}} \otimes \mathbf{E}_3 + \mathbf{E}_2 \otimes \mathcal{H}_{\text{trip}} + \\ & \begin{pmatrix} \frac{D_{zz}+J}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{3}}\left(-\frac{D_{zz}}{2} + J\right) & 0 & 0 \\ 0 & 0 & -\frac{D_{zz}+J}{2} & 0 & \frac{1}{\sqrt{3}}\left(-\frac{D_{zz}}{2} + J\right) & 0 \\ 0 & \frac{1}{\sqrt{3}}\left(-\frac{D_{zz}}{2} + J\right) & 0 & -\frac{D_{zz}+J}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}}\left(-\frac{D_{zz}}{2} + J\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{D_{zz}+J}{2} \end{pmatrix} \end{aligned} \tag{31}$$

$$\hat{\mathcal{H}}_{\text{commutator superoperator}} = \mathbf{E} \otimes \mathcal{H} - \mathcal{H}^{\dagger} \otimes \mathbf{E} \tag{32}$$

5.1.2 Relaxation

The relaxation superoperator $\hat{\hat{\mathbf{R}}}$ is a square matrix with the squared dimension of the system. The phenomenological superoperator with the longitudinal relaxation rate $k_1 = 1/T_1$ and the transverse relaxation rate $k_2 = 1/T_2$ is set up as follows:

$$\begin{aligned} \hat{\hat{\mathbf{R}}}_{ii,jj} &= k_1 \\ \hat{\hat{\mathbf{R}}}_{ij,ij} &= k_2 \\ \hat{\hat{\mathbf{R}}}_{ii,ii} &= -\sum \hat{\hat{\mathbf{R}}}_{ii,jj}. \end{aligned} \tag{33}$$

As an example, the matrix is given for a spin 1 system:

$$\hat{\hat{\mathbf{R}}} = \begin{pmatrix} -2k_1 & 0 & 0 & 0 & k_1 & 0 & 0 & 0 & k_1 \\ 0 & k_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_2 & 0 & 0 & 0 & 0 & 0 \\ k_1 & 0 & 0 & 0 & -2k_1 & 0 & 0 & 0 & k_1 \\ 0 & 0 & 0 & 0 & 0 & k_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & k_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_2 & 0 \\ k_1 & 0 & 0 & 0 & k_1 & 0 & 0 & 0 & -2k_1 \end{pmatrix}. \tag{34}$$

5.2 Simulation example

Listing 1: Source code for the simulation of the spin system described in the main part. The four steps to the full spectrum are marked.

```
# %% Step 1: Determination of the spin system
```

```python
class Sys:
    pass

class Exp:
    pass

class SimOpt:
    pass

# choose a triplet doublet pair spin system
Sys.spin_system = "tdp"

# set up g-tensors of the radical and the triplet
Sys.g = [2.0103, 2.0070, 2.0025]
Sys.g_tri = [2.00522, 2.00522, 2.00522]

# define triplet ZFS
Sys.D_tri = -1681
Sys.E_tri = 40

# define couplings of the radical and the triplet
Sys.J_ex = 400000
Sys.D = -660

# define initial state
Sys.precursor = "triplet-zf"
Sys.population = [0.53,  0.60, 0.17, 0.14, 0.46]

# define decay time and line width
Sys.decay = 3e-6
Sys.width_gauss = 5

# experimental setup
Exp.B_z = np.linspace(271.492, 432.075, 1000)
Exp.t_scale = [0, 1e-6]
Exp.t_points = 2
Exp.B_mw = 0.001
Exp.freq_mw = 9.75e9

# simulation options
SimOpt.grid = "fibonacci"
SimOpt.grid_points = 14
SimOpt.space = "hilbert"
SimOpt.eigval_mode = False

# do the simulation
spc = teacups(Sys, Exp, SimOpt)


# %% Step 2: Analysis of the energy of the eigenstates

SimOpt.eigval_mode = True
eigenvalues = teacups(Sys, Exp, SimOpt)


# %% Step 3: Set up the dynamics

k_ic = 7/1e-6
k_isc = 1.5/1e-6

R = np.zeros((6, 6))

# Doublet
R[0, 1] = k_ic
R[1, 0] = k_ic

# +- 3/2
R[2, 5] = k_ic*2.5
R[5, 2] = k_ic*2.5

# +- 1/2
R[3, 4] = k_ic
R[4, 3] = k_ic

# +-3/2 <-> +- 1/2
R[2, 3] = k_isc
R[3, 2] = k_isc

R[4, 5] = k_isc
R[5, 4] = k_isc

Sys.dynamics = R

# Alternative for phenomenological relaxation superoperator
# Sys.T_relax_1 = 1e-6
# Sys.T_relax_2 = 1e-6
```

```
# %% Liouville-space simulation

SimOpt.eigval_mode = False
SimOpt.space = "liouville"
SimOpt.pop_evolution = True
Exp.t_points = 100

# Run the simulation
spc = teacups(Sys, Exp, Opt)
population_evolution_array = Exp.pop_evolution
```

5.3 Programming details

The whole code of `teacups` can be found at URL. Some details about the programming are given below. This can be used as a short overview of the code and an introduction to the main documentation. It helps finding a starting point. The full documentation can be found at URL.

`teacups` consists of 13 Python modules, which contain classes and functions. They can be roughly sorted into three categories, namely specific simulation modules, general modules, and structuring modules.

5.3.1 Specific simulation modules

These modules include all functions that are specific for the simulation of transient EPR spectra:

- creators.py
- density_matrices.py
- hamiltonians.py
- hyperfine.py
- input_handler.py
- relaxation.py
- signals_and_processing.py
- simulations.py.

The main file simulations.py calls all functions in the right order to obtain the wanted spectrum. The other modules are used to set up and transform the explicit matrices as described in the theory part. The functions of the specific simulation modules use a four-class system as input and output, which consists of `Sys`, `Exp`, `SimOpt` and `Cal`. All information needed for the simulation is provided and sorted in these four classes as attributes:

- `Sys` contains all spin system parameters including relaxation times.
- `Exp` contains all experimental parameters.
- `Opt` contains simulations options, e.g. the choice of the space, the number of grid points, and the simulation mode.
- `Cal` is at the beginning an empty class, but it is filled with the results of the calculations during the simulation. These results are fed to further functions later.

5.3.2 General modules

These modules contain functions that are generally usable (and could be used outside of `teacups`):

- convolution.py
- grid.py
- orientation_dependent_ham.py
- relaxation.py.

Included are, amongst other things, the setup of a grid on a sphere, the Gaussian convolution of a spectrum, or the general setup of a relaxation matrix/ Hamiltonian. The functions take parameters as input and return the desired output. They do not use the four-class system, which makes them more flexible.

5.3.3 Structuring modules

All matrices of the specific simulation modules are objects of the classes defined in the structuring modules:

- matrix_tools.py
- multioperator_tools.py.

Both classes define matrices including methods like rotation to a number of angle combinations, the setup of a Hamiltonian etc. An object of class `multioperator` has an attribute called `B_angle_matrix`. This is a three-dimensional `NumPy` array containing the values of the operator for a number of magnetic field points and a number of angle points. Its dimension is `len(B)` $\times$ `len(angles)` $\times$ **`spinsystem dimension`** $\times$ **`spinsystem dimension`**. The specific simulation modules often use objects of multioperator-type, which has the advantage, that calculations can be done for all field points and angles at once without using loops. This makes the code much faster.

5.4 Code optimisation

The calculation of the time-evolution of an experimental spectrum is computationally rather expensive. Thus, some effort has been invested to optimise the performance of the code. In the following, an overview of the optimisation attempts is given:

- **Vectorisation using NumPy:** Using the multioperator-class (as described above) all calculations are performed simultaneously since the matrices are collected in one big array. This has great advantages as nearly no loops are needed and fast built-in NumPy functions (partly based on efficient Fortran routines from LAPACK) are used. This comes with the disadvantage that more memory is needed to save the big NumPy-arrays.
- **Parallel computing:** The NumPy functions automatically use multiple cores, which works well. For future improvement, parallel computing may be applied to the time-evolution loop. The array could be split along the magnetic field axis and the evolution for multiple magnetic field points could be done simultaneously.
- **Numba:** Using just-in-time-compilation with Numba could increase the performance of Python-loops. As most of the code is vectorised, no loops are needed and Numba could only be used for the time-evolution loop. It should be noted that Numba does not work in combination with SciPy (see below).
- **The matrix exponential:** The bottleneck of the simulations is the calculation of the matrix exponential. Using NumPy, it can be calculated by diagonalising the exponents matrix. For this purpose, the function numpy.eig is used. The faster function numpy.eigh cannot be used as the matrix is not Hermitian. NumPy uses Fortran LAPACK routines for the diagonalisation and already works rather efficiently. Nevertheless, we tried to speed up the calculations.
- **SciPy:** For the calculation of the matrix exponential, the SciPy function works more efficiently than the NumPy function. However, using SciPy, vectorisation cannot be used and loops are needed for magnetic field points and angle points. The latter cannot be accelerated using Numba as Numba does not work in combination with SciPy. Benchmarks showed that the use of NumPy without loops is faster than the use of SciPy with loops.
- **Julia:** As the loops seemed to be the problem when using SciPy, Julia-code has been integrated for the calculation of the matrix exponential. Due to just-in-time-compilation, loop-overlays are faster in Julia than in Python. Nevertheless, the performance could not be optimised since the matrix exponential function in Julia also uses the Fortran LAPACK routine. Consequently, the calculation was found to be just as fast using Julia as compared to NumPy.
- **FastExpm and Expokit:** In Julia, some packages for the fast calculation of matrix exponentials are available. However, the maximum matrix size of 36x36 for a triplet-douplet pair in Liouville space is still to small to achieve a significant improvement in performance using these functions.
- **Using the GPU:** For python the CUDA package may be used to calculate expensive matrix algebra on the GPU. Until now no function for non-hermitean matrix-exponentials exists but as soon as it does calculations on the GPU could significantly fasten the code.